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Impact of high pressure on high concentration carbon dioxide capture from natural gas by monoethanolamine/*N*-methyl-2-pyrrolidone solvent in absorption packed column

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ABSTRACT

Carbon dioxide (CO_2) removal is an essential step in natural gas (NG) processing to provide high quality gas stream products and minimize operational difficulties. This study aims to investigate the absorption performance of CO_2 at high concentration level of 50% from a mixture of CO_2–NG gas stream at pressure condition of up to 5 MPa for 20 wt% monoethanolamine (MEA) aqueous solution and MEA hybrid solution consisting of 20 wt% MEA with 40 wt% *N*-methyl-2-pyrrolidone (NMP) + 40 wt% water in a countercurrent packed column. Comparison of absorption performance was also conducted with water and NMP in order to understand the effect of physical absorption by the individual solvent respectively. It was observed that water had minimal physical absorption for CO_2. On the contrary, NMP was observed to be an efficient physical absorbent for CO_2 on its own. Increased pressure was found to have a positive impact on the CO_2 removal performance for water, NMP, MEA aqueous solution and MEA hybrid solution. This trend was especially significant when the pressure was at 5 MPa.

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1. Introduction

CO_2 removal is an essential process in natural gas sweetening. Raw natural gas typically contains a large quantity of methane along with heavier hydrocarbons such as ethane, propane, isobutene, *n*-butane and significant amount of CO_2 (Faiz and Al-Marzouqi, 2011). Prior to sale or usage, CO_2 must be removed to an acceptable level, as CO_2 is highly corrosive in the presence of moisture thus rapidly destroys pipelines and equipment; it also reduces the heating value of a natural gas stream and wastes pipeline capacity (Safari et al., 2009).

Natural gas in commercial operations constitute varying amounts of CO_2 ranging from CO_2–free natural gas in Siberia to CO_2 content of up to 90% in the Platong and Erawan fields in Thailand. The Natuna field in the Greater Sarawak Basin in Indonesia is the largest gas field in Southeast Asia, with an estimated 46 trillion cubic feet of recoverable reserves. Unfortunately, it remains unexplored primarily due to high CO_2 content at 71% (IEA, 2008a). Meanwhile, natural gas reserves with low CO_2 content are rapidly depleting with the continuous extraction of the fossil fuel over the years. Hence, there is an urgent need to explore the high CO_2 content natural gas reserves in order to sustain the power requirement by human population worldwide (Mak et al., 2003). Meanwhile, in steam reforming of natural gas, it is also important to separate CO_2 from H_2 in the pre-combustion capture process before H_2 could be fed into combustion turbine (Olajire, 2010).

In Malaysia, CO_2 content in natural gas fields varies from 28 to 87%. Over 13 trillion cubic feet of natural gas reserves are undeveloped due to high concentration of CO_2 (Darman and Harun, 2006). These high CO_2 content gas fields in Malaysia represent significant CO_2 capture and storage (CCS) opportunity. Besides that, pre-combustion CO_2 capture i.e. gasification plants with high pressure and high CO_2 concentration synthesis gas is also another area of opportunity for CCS (Figueroa et al., 2008).

Globally, as of year 2011, there were 62 active or planned commercial scale integrated CCS projects, comprising capture, transport and sequestration elements, for sequestering of CO_2. However, only seven projects were in the operational stage while the rest were in the evaluation, definition, or execution stages (Hardisty et al., 2011). As such, valuable insight generated by continuous and intense research on CCS is required in order to accelerate the transition from evaluation to operational stage.

In the gas processing industry, cryogenic separation is usually used for high concentration CO_2 (typically more than 50%) purification from gas stream. The advantage of cryogenic separation is that it enables direct production and transport of liquid CO_2 (Gupta

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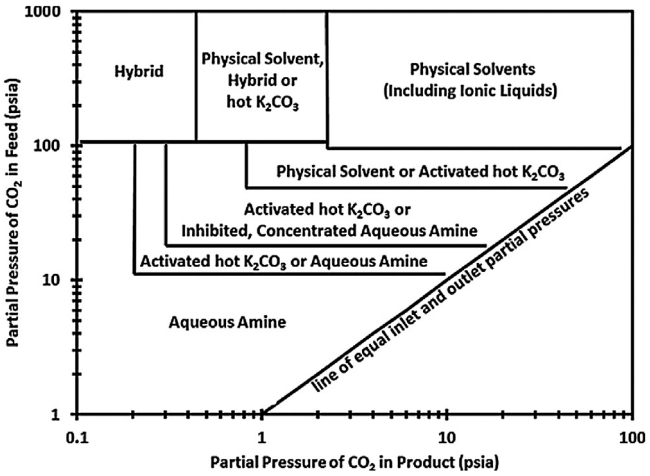


Fig. 1. Process selection guidelines for CO_2 removal using solvent based process (Shannon et al., 2011).

et al., 2003). ExxonMobil developed a single step cryogenic distillation process called Controlled Freeze Zone™ (CFZ) process in year 1986 aimed for handle high CO_2 and H_2S gas reserves. It was capable of handling natural gas with a wide range of CO_2 from 15% to 65% (Northrop and Valencia, 2009). This gas treatment technology has been recently deployed in the Shute Creek plant, USA as commercial demonstration facility. The success of this commercial demonstration would enable the exploration of high CO_2 content natural gas reserves around the world (Michael et al., 2011).

Meanwhile, amine systems have been widely used for many years to remove entrained CO_2 in raw natural gas (Abu-Zahra et al., 2007; Hardisty et al., 2011). One of the advantages of amine systems is low hydrocarbon loss, besides being able to meet the low CO_2 specification required in processed natural gas (Sridhar et al., 2007). In general, most of the existing CO_2 capture systems are based on chemical absorption combined with heat induced CO_2 recovery (Desideri, 2010; IEA, 2008a).

For elevated pressure absorption system, water was formerly used for CO_2 absorption but due to its low CO_2 removal efficiency and extremely high pumping load, it was gradually replaced with more efficient system using chemical or organic physical solvents (Kohl and Nielsen, 1997). However, the chemical solvents are usually diluted in water before usage under continuous absorption process in order to reduce the chemical's corrosiveness. Among the solvents available, the most widely used solvent in industrial process for capturing CO_2 is monoethanolamine (MEA), which is predominantly diluted to 10–30 wt% in water (Aroonwilas and Tontiwachwuthikul, 1997; Dave et al., 2009; Sønderby et al., 2013).

Absorption in a physical solvent relies on the solubility of CO_2 in the solvent rather than the chemical reaction with the solvent. It is usually more suitable for high pressure bulk removal application (Chakma, 1999). The solvent could be regenerated by varying the pressure or temperature. However, since the CO_2 concentration in flue gas from coal fired power plant is typically up to 15% (Dave et al., 2009; IEA, 2008a; Thomas and Kerr, 2005; Yan et al., 2008) at low pressure condition, chemical absorption is generally preferred for CO_2 capture in flue gas because of their higher absorption capacity at low CO_2 partial pressure (Aroonwilas and Tontiwachwuthikul, 1997; Schwendig, 2010).

Hybrid solvent, which consists of both physical and chemical solvents, has the potential to minimize the weakness of each solvent whilst simultaneously offer the opportunity to formulate the solvent according to the high pressure offshore field or onshore gas received condition. As indicated in Fig. 1, hybrid solvent is expected to provide fine cleanup at high CO_2 partial pressure.

To date, information on hybrid solvent is very limited in open literature. In recent years, there are growing research on hybrid solvents using a combination of methanol with diethanolamine (DEA) or MEA (Archane et al., 2008; Usubharatana and Tontiwachwuthikul, 2009) and *N*-methyl-2-pyrrolidone (NMP) with diethylene glycol (DEG) and MEA (Vaidya and Mahajani, 2005). However, thus far, these studies were conducted under batch or semi-batch condition.

von Harbou et al. (2014) assessed the potential of using ethanol as co-solvent for aqueous MEA based on simulations with an equilibrium stage model for post-combustion CO_2 capture process and found that the absorption rate of CO_2 was enhanced by addition of ethanol. However, its volatility caused very high ethanol loss through the top of the absorber. This led to increased complexity of the capture plant due to incorporation of an additional ethanol recovery unit. As such, it was concluded that other solvents bearing similar properties yet have lower vapor pressure should be used instead.

Usubharatana and Tontiwachwuthikul (2009) who studied on the enhancement factor and kinetics of CO_2 capture using MEA–methanol hybrid solvent under batch condition found that when methanol was mixed into the MEA solution, the rate of MEA deprotonation decreased. However, the CO_2 absorption rate increased with the presence of methanol in the MEA hybrid solvent as methanol enhanced the diffusivity and solubility characteristics of the solvent. This showed the complex behavior of hybrid solvent; hence, more studies need to be conducted for each combination of hybrid solvent in order to understand its unique behavior.

Due to the increasing need to explore the high pressure and high CO_2 content natural gas fields in order to fulfill the rising energy demand worldwide (IEA, 2008b), more efficient solvents should be developed to purify natural gas with high CO_2 content. Mixing a specific organic solvent into chemical solvent is expected to provide fine cleanup for natural gas processing due to improved affinity to absorb CO_2 compared to water. However, a comparison study on the absorption performance of physical solvent, aqueous chemical solvent and hybrid solvent under continuous flow process at high CO_2 concentration and high pressure condition needs to be conducted to establish better understanding of the hybrid solvents unique characteristics. Hence, the main objective of this work is to study the absorption performance of high CO_2 concentration natural gas at various pressure conditions in a countercurrent packed column using a solvent which consists of MEA as well as water and/or NMP. A study on the individual physical solvents' performance was also conducted in order to gain insight on its ability to absorb CO_2 on its own. NMP was selected as the organic physical solvent in this study due to the following reasons:

(a) Low viscosity and is completely miscible with water. This would facilitate pumping and reduce pumping power needed for the process (Shannon et al., 2011).
(b) Able to tolerate and remove H_2S. It could be used for treating high pressure gases with high sour gas concentrations at ambient temperature (Hochgesand, 1970; Shannon et al., 2011). This means that it could be applied for offshore application which deals with significant amount of H_2S.
(c) Its maximum operating temperature is high (200 °C). This would enable operation at a wide temperature range (Shannon et al., 2011).

NMP is the physical solvent used in the Purisol process due to its attractive properties of low vapour pressure, low viscosity and high CO_2 absorption capacity (Tian et al., 2012). Being a very polar organic solvent, NMP was found to have superior CO_2 solubility compared to dimethylsulfoxide (DMSO) at elevated pressure up to 9 MPa (Rajasingam et al., 2004).

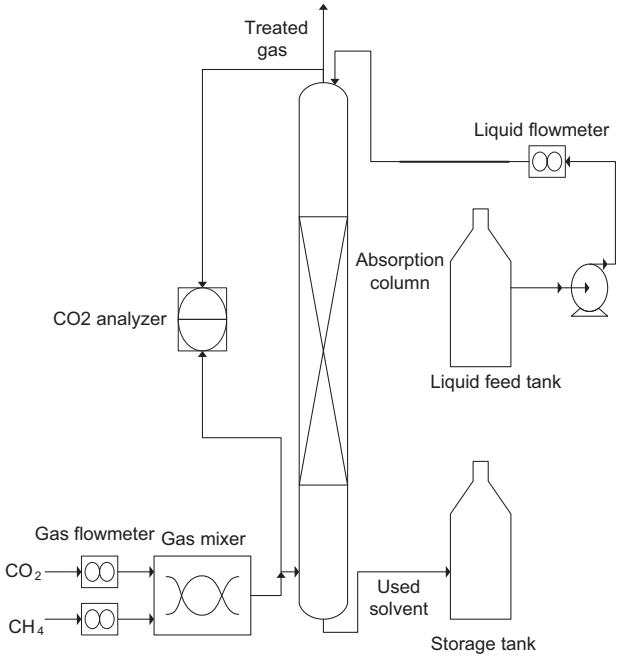


Fig. 2. Schematic diagram of experimental setup.

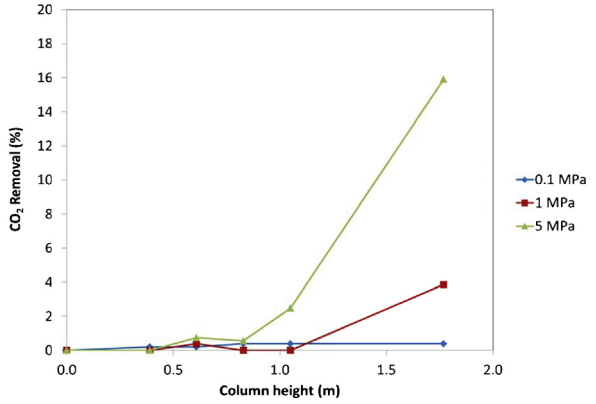


Fig. 3. CO_2 removal performance along the absorption column at steady state condition for water.

However, experimental data on the absorption performance of NMP in countercurrent packed column for high concentration of CO_2 (50%) in feed gas has not been presented in open literature to date. This paper would provide insight on the impact of high pressure (up to 5 MPa) to the MEA aqueous and MEA–NMP hybrid solution for CO_2 absorption performance in packed column.

2. Materials and methods

2.1. Materials

Monoethanolamine (MEA) (99%) and *N*-methyl-2-pyrrolidone (NMP) (99%) used in this study were purchased from Acros Organics. Distilled water was used to prepare the aqueous and hybrid solvent for this work. CO_2 (99.9%) used in this experiment was purchased from Air Product Malaysia while natural gas with 97% of CH_4 and 2% of CO_2 was supplied by PETRONAS Dagangan Bhd.

2.2. Experimental set up and procedure

A schematic diagram of the experimental setup is shown in Fig. 2. The absorption column has an inner diameter (ID) of 0.145 m and packing height of 1.0 m. The packing used in this study was structured packing FLEXIPAC 1Y by Koch-Glitsch. There was a distance of 0.5 m between the packing and liquid distributor.

The absorption experiment was initiated by introducing a gas mixture of CO_2 and natural gas to the bottom of the absorption column. The concentration of CO_2 in the CO_2/NG feed gas mixture was adjusted using the mass flow controller. A back pressure regulator was installed to maintain the pressure of the absorption column at the desired level. Liquid feed was then pumped to the top of the column at a flow rate of 1 l/min from a 200-l feed tank to create countercurrent contact between gas and liquid. The absorption process was operated until steady state condition was achieved.

The CO_2 concentration in the feed gas and treated gas was measured from the start of the experiment until it reached steady state, using CO_2 infrared (IR) gas analyzer model ZRJF4Y25 (Fuji Electric Instruments). There were four sampling points along the height of the packing. The CO_2 removal efficiency (η) which defines the percentage of CO_2 in the gas stream that was removed during the absorption operation was determined from the difference between the amounts of CO_2 entering and leaving the column, and can be expressed by the following equation:

$\eta = \left[\frac{(y_i - y_o)}{y_i}\right] \times 100\%$ where y_i and y_o denote percentage of volume concentration of CO_2 entering and leaving the absorption column at steady state, respectively (Yeh et al., 2001).

Specifically, this work studied the effect of pressure on CO_2 removal by water, NMP, 20 wt% MEA aqueous solution and 20 wt% MEA with 40 wt% NMP + 40 wt% water. The column pressure was varied between 0.1 MPa and 5 MPa, CO_2 inlet concentration was set at 50% and the total gas flow rate was 100 slpm. The solvent was pumped into the column without heating. For this work, the temperature of water and NMP was recorded to be at approximately 30 °C while the temperature of the two MEA solutions was at approximately 35 °C.

3. Results and discussion

3.1. Effect of pressure on physical solvents

Fig. 3 shows the CO_2 removal performance along the column at steady state condition using water as the physical solvent. T "zero" column height represents the location of CO_2 inlet while the highest point was the location of liquid distributor where fresh solvent and gas first came into contact. Water was included as a physical solvent in this work as the reference point. Since it would also be present in MEA solution, it was necessary to understand the contribution of water in removal of CO_2 absorption physically. From the graph, it was observed that the CO_2 removal efficiency improved with increasing pressure. Less than 1% of CO_2 was removed by water at 0.1 MPa. At 1 MPa, slight improvement was observed as approximately 4% of CO_2 was removed at the gas outlet. CO_2 removal efficiency further improved when the column pressure was increased to 5 MPa. The results obtained were consistent with previous report that CO_2 was relatively insoluble in water due to liquid film resistance (Kohl and Nielsen, 1997). Kohl and Nielsen (1997) reported that water had low CO_2 removal efficiency even at elevated pressure.

Fig. 4 depicts the CO_2 removal performance along the column at steady state condition using NMP as the organic physical solvent. The results obtained showed 100% CO_2 removal at gas outlet at all three pressure condition tested i.e. 0.1 MPa, 1 MPa and 5 MPa. This showed that NMP was an efficient physical absorbent since CO_2 was completely removed even at a very short column distance; 0.39 m at both 1 MPa and 5 MPa as well as 0.83 m at 0.1 MPa.

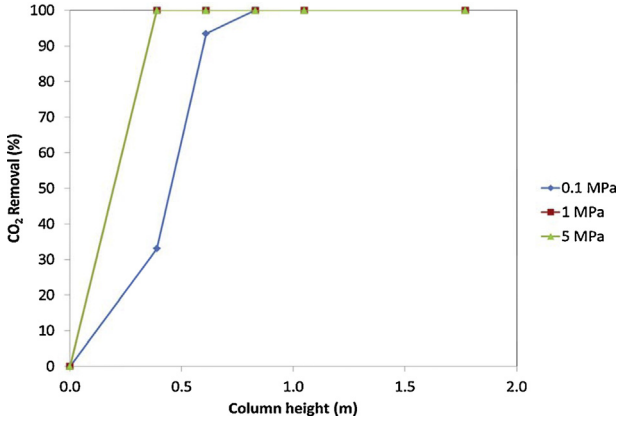


Fig. 4. CO_2 removal performance along the absorption column at steady state condition for NMP.

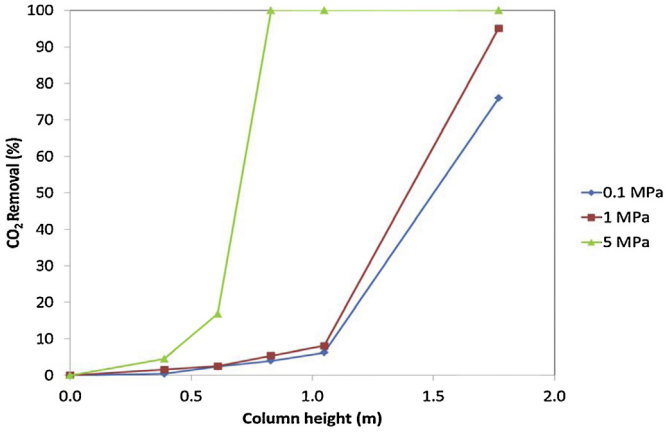


Fig. 5. CO_2 removal performance along the absorption column at steady state condition for 20 wt% MEA aqueous solution.

3.2. Effect of pressure on MEA solution

Fig. 5 shows the CO_2 removal performance along the column at steady state condition for 20 wt% MEA aqueous solution. Approximately 76% of CO_2 removal was achieved at the gas outlet when the pressure was low (0.1 MPa). The removal efficiency was observed to be improved to 95% when the column pressure was 1 MPa. This trend was consistent with findings by Godini and Mowla (2008) who also observed increased CO_2 removal efficiency when CO_2 absorption test was performed in a pilot plant using MEA aqueous solution at pressure condition between 0.1 and 1 MPa. Zhang et al. (2003) who conducted a CO_2 absorption study in packed column using MDEA and piperazine (PZ) up to 0.625 MPa also indicated that the absorption rate of CO_2 at elevated pressure was larger than that of atmospheric pressure because of the Marangoni effect. The Marangoni effect could occur because at high CO_2 partial pressure, there was an increase of interfacial tension resulting from the concentration increase in the interface. This condition would increase the instability of the gas/liquid-phase interface and lead to interfacial disturbance (Tan et al., 2012).

Based on Fig. 5, CO_2 was completely removed at an elevated pressure of 5 MPa at the gas outlet. In fact, the CO_2 absorption performance was observed to be significantly improved at high pressure as compared to CO_2 absorption at 1 MPa because 100% CO_2 removal was detected from a column distance of 0.83 m onwards. Despite the earlier observation from Fig. 3 in Section 3.1 that the increase of pressure from 1 MPa to 5 MPa only had minimal influence to the CO_2 removal performance in the middle of the column

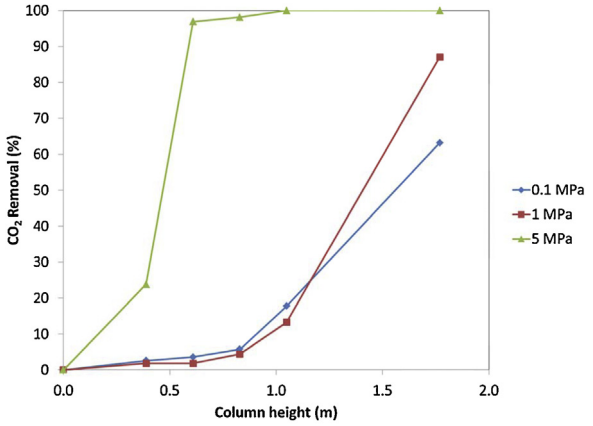


Fig. 6. CO_2 removal performance along the absorption column at steady state condition for 20 wt% MEA solution with 40 wt% NMP + 40 wt% water.

when water was used as the solvent, results shown in Fig. 5 indicated that at high pressure of 5 MPa, the CO_2 intake capacity of the MEA aqueous solution could be substantially enhanced. This observation was in line with finding by Ghulam et al. (2013) that the CO_2 solubility of amine solution was significantly improved at high pressure of up to 6 MPa.

From Fig. 5, it was also observed that at 0.1 MPa and 1 MPa, substantial difference of CO_2 removal performance was detected from the point of column height of 1.77 m to 1.05 m. This was because at 1.77 m, fresh solvent first came into contact with the gas stream in the column. At this point, the active absorbent in the fresh solvent has the highest absorption capacity; hence, the efficiency of CO_2 removal was highest. At 1.05 m column height, the solvent had reacted with CO_2 to a certain extent. Hence, the CO_2 loading in the solvent which increased as the solvent travelled downwards from height of 1.77 m to 1.05 m in this countercurrent absorption process reduced the efficiency of CO_2 removal as observed in Fig. 5.

Increase of CO_2 loading in the solvent reflected lower available active absorbent concentration in the solution and this condition could significantly reduce the mass transfer coefficient (Buzek et al., 1997). Arashi et al. (1997) observed that when CO_2 loading in the lean solution was high, the driving force of mass transfer from gas to solution became small and the decrease in CO_2 removal efficiency could not be compensated even with the increase of solution flow rate. Similar trend was found by Dinca and Badea (2013) whereby the CO_2 removal capacity was significantly reduced in rich CO_2 loading MEA solvent. The strong influence of CO_2 loading to the mass transfer coefficient was also discovered by Setameteekul et al. (2008). Among the process parameters studied (amine concentration, CO_2 partial pressure in feed gas, temperature, CO_2 loading of solvent and liquid circulating rate), the authors observed that CO_2 loading of the solution was the most influential process parameter affecting the overall mass-transfer coefficient.

Fig. 6 depicts the CO_2 removal performance along the column at steady state condition for hybrid solution consisting of 20 wt% MEA solution with 40 wt% NMP + 40 wt% water. A similar trend was also observed for this hybrid solution with substantial improvement of CO_2 removal at 5 MPa.

3.3. Comparison of CO_2 removal efficiency among the solvents

Comparison of CO_2 removal performance along the column at steady state for NMP, 20 wt% MEA aqueous solution and 20 wt% MEA solution with 40 wt% NMP + 40 wt% water at 0.1 MPa, 1 MPa and 5 MPa is shown in Figs. 7–9, respectively. Generally, the trend of CO_2 removal along the column for 0.1 MPa and 1 MPa was similar, with NMP showing the highest CO_2 removal efficiency, followed

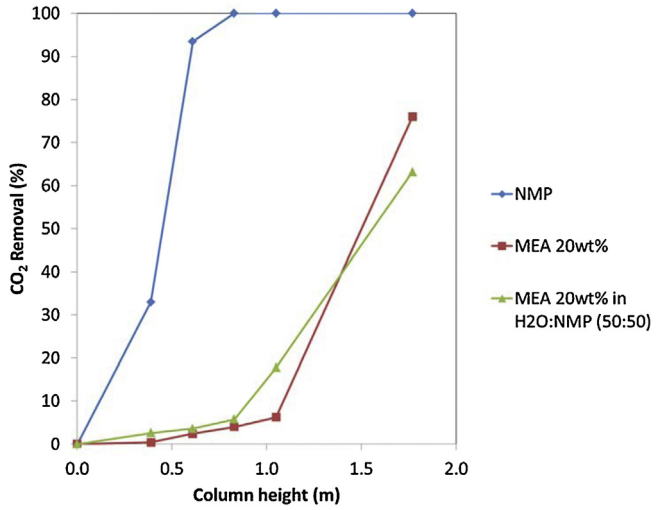


Fig. 7. Comparison of CO_2 removal performance along the absorption column at steady state condition at pressure of 0.1 MPa.

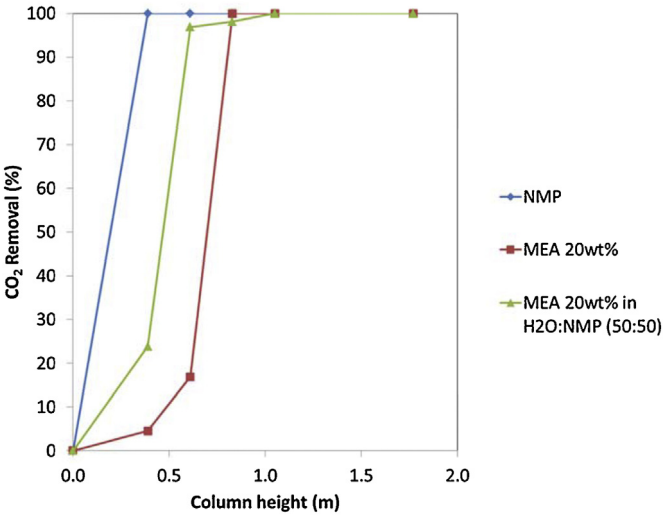


Fig. 9. Comparison of CO_2 removal performance along the absorption column at steady state condition at pressure of 5 MPa.

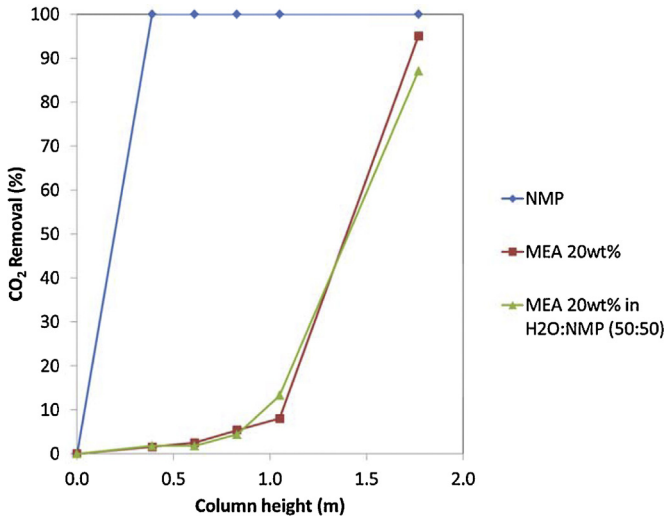


Fig. 8. Comparison of CO_2 removal performance along the absorption column at steady state condition at pressure of 1 MPa.

by MEA aqueous solution and MEA hybrid solution. The results seemed to be in agreement with observation by Rivas and Prausnitz (1979) who also found that NMP was the best solvent for absorbing CO_2 as the Henry's constant for NMP was the lowest among the five solvents tested (i.e. monoethanolamine, diglycolamine, sulfolane, propylene carbonate and NMP) at pressure below 1 atm. The seemingly superior performance of NMP, even at low pressure, was surprising since the typical equilibrium lines for chemical and physical absorption indicated that physical absorption is favored at high CO_2 partial pressure while chemical absorption is favored at low CO_2 partial pressure (Hochgesand, 1970).

However, Vaidya and Mahajani (2005), who studied the reaction kinetics of CO_2 with MEA + NMP + DEG (diethylene glycol) using stirred cell reactor operated batch-wise, investigated the role of NMP in the hybrid solvent by conducting two experiments in the absence of MEA at low CO_2 partial pressure up to 5 kPa. They observed that the rate of CO_2 absorption was higher than that expected. They deduced that in addition to physical absorption, chemical reaction between CO_2 and NMP could possibly take place.

From Figs. 7 and 8, it was observed that the CO_2 removal performance for the hybrid solution at gas outlet for 0.1 MPa was only approximately 63% if compared to 76% removal for MEA aqueous solution. The CO_2 removal efficiency at gas outlet for the hybrid solution at 1 MPa was also lower (87%) compared to MEA aqueous solution (95%). Leites (1998) compared the CO_2 solubility in aqueous MEA and MEA–organic solvents mixtures at CO_2 partial pressure of up to 0.103 MPa and found that CO_2 solubility in MEA aqueous solution was higher compared to solubility in MEA–NMP solution. However, the difference between them was not significant. Therefore, the CO_2 removal performance trend observed for MEA aqueous and hybrid solution at lower pressure seemed to be in agreement with the observation on CO_2 solubility by Leites (1998).

On the other hand, Fig. 9 shows that all three solvents achieved 100% CO_2 removal at the gas outlet for at 5 MPa. Substantial increase of CO_2 absorption along the column suggested increased CO_2 intake capacity at higher pressure. This higher intake is expected as carbamate hydrolysis can occur with amine, particularly at high pressures. This hydrolysis would generate free amine which can react with additional CO_2, hence increasing the loading of the amine solution (Sartori and Savage, 1983).

Based on the solubility study for partial pressure of CO_2 up to 0.005 MPa, Vaidya and Mahajani (2005) observed that NMP enhanced the solubility of CO_2 in hybrid solution. It was also deduced that the addition of NMP did not interfere with the reaction between MEA and water (Vaidya and Mahajani, 2005). However, based on Figs. 7–9, when a portion of the water was replaced with NMP, the higher CO_2 intake capacity trend was only observed for MEA hybrid solution at 5 MPa. At 0.1 MPa and 1 MPa, it was observed that MEA aqueous solution seemed to have better CO_2 removal performance compared to MEA hybrid solution. This suggested that different absorption mechanism might be taking place for the hybrid solution when it was at high pressure condition.

The results from Figs. 8 and 9 were consistent with findings by Rivas and Prausnitz (1979) who conducted preliminary design calculations for sweetening of sour natural gas based on gas solubility data. They found that at 1.5 MPa, the flow rate for MEA–NMP solvent had to be 50% more than the flow rate for MEA aqueous solution in order to achieve similar CO_2 removal performance. However, when the pressure was increased to 7 MPa, MEA–NMP solvent could achieve similar CO_2 removal performance with aqueous MEA at lower MEA–NMP solvent flow rate. According to their study, this was related to the solubility of CO_2 in the respective

solvents whereby MEA–NMP was found to be inferior compared to aqueous MEA at lower CO_2 partial pressure but the opposite was observed when the CO_2 partial pressure continued to increase.

4. Conclusions

From this study, it was confirmed that water had minimal physical absorption for CO_2. On the contrary, NMP was observed to be an efficient physical absorbent for CO_2. Increasing pressure was found to have positive impact on CO_2 removal performance for water, NMP, MEA aqueous solution and MEA hybrid solution. This trend was especially significant when the pressure was increased to 5 MPa. The CO_2 intake capacity for MEA solutions were noticeably increased at high pressure condition. The results from this study suggested that high pressure could accelerate the reaction between MEA aqueous solution and MEA hybrid solution with CO_2. The MEA–NMP hybrid solution showed better CO_2 removal performance compared to the aqueous solution only at 5 MPa. At 0.1 MPa and 1 MPa, it was observed that MEA aqueous solution had better CO_2 removal performance compared to MEA hybrid solution.

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